Confounders and controls

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which other variable moves with the intended factor and what comparison could separate it.

Teach explicit controls, repeated-view identity and bounded diagnostic claims.

Keep one habit: Model multi-change comparison → cross leaf/background cells → inspect nuisance-only rule → assess timing counterbalance → retain unrecorded-condition limits.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **confounder**s and **control**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of experimental units and independence

Reference example, not the only acceptable wording: Teacher explanation: The experimental unit is the independently assigned or sampled entity. Multiple measurements from one unit are often related.

If methods are compared within each query, queries are paired units. Repeated runs within a query estimate run variation but do not increase the number of independent query contexts.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of hypotheses and measurable outcomes

Reference example, not the only acceptable wording: Teacher explanation: A hypothesis predicts a measurable comparison under stated conditions. Declare primary outcome and direction before inspecting results.

Toy hypothesis:method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of methods results and limitations

Reference example, not the only acceptable wording: Teacher explanation: Methods state what was done; results state observed outcomes; limitations state restrictions on interpretation. Mixing them hides missing **evidence**.

Classify each statement by role and link results to the procedure and denominator. A proposed **mechanism** is an explanation to test, not an extra **observation**.

Pictured end goal

Explain **confounder**s and **control**s using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Experimental units and independence

Hypotheses and measurable outcomes

Methods results and limitations

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A **confounder** changes alongside the intended factor and can offer another explanation. **Control**s aim to isolate the proposed **mechanism**.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **confounder**s and **control**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach controls as explicit comparisons that challenge alternatives

> Keep metadata checks, diagnostic test designs and actual causal evidence distinct.

Definitions and complete fictional control contract

A **confounder** changes alongside the intended factor and can offer another explanation. **Control**s aim to isolate the proposed **mechanism**.

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Control-plan item	Complete fictional classroom contract
Intended main factor	Retrieval method A versus B
Main unwanted changes	A uses corpus c1 and Q1–Q3; B uses c2 and Q4–Q6
Fixed declared fields	Cutoff 3, rubric r1 and toy-local environment
Proposed matched comparison	B uses c1 and Q1–Q3 as A does, with method B retained
Evidence gap	No retrieval outcomes or real rerun are supplied; matching metadata is a plan/check, not a performance result
Separate picture factor	Leaf condition healthy/spotted; nuisance background white/blue

Control-plan item	Complete fictional classroom contract
Original coupled cells	Healthy on white; spotted on blue only
Crossed design	Healthy-white, healthy-blue, spotted-white, spotted-blue
Scope	Fictional design metadata and an explicit toy paper-only rule; no actual images, collection, classifier training or causal experiment

Main intended-change ledger and checked reasoning

Added main metadata field	A original condition	B original condition	Comparison interpretation
Method	A	B	Intended difference
Corpus	c1	c2	Unwanted difference for isolated method question
Queries	Q1, Q2, Q3	Q4, Q5, Q6	Unwanted case difference
Cutoff/rubric/environment	3 / r1 / toy-local	3 / r1 / toy-local	Declared matches, not independent execution verification
Proposed repair	Keep A as stated	Method B with c1/Q1–Q3/3/r1/toy-local	Matched-plan proposal; no new score supplied
Remaining evidence	Actual outcomes, order and further conditions	Still need to be collected/recorded	Do not claim all confounding solved

Main A/B changes corpus and queries as well as method, so an isolated method claim is unsupported. Match corpus, queries, cutoff, rubric and environment; handle order effects and record actual conditions. The local checker only inspects supplied fields. Picture crosses healthy/spotted with white/blue.

Explicit paper-only rule fits original 2/2 but crossed 2/4; this demonstrates a possible nuisance explanation, not actual trained-model reasoning. Same H1/S1 across backgrounds are repeated views, and all views must remain held out for a new-object test. More healthy-white/spotted-blue alone preserves confounding.

Selected A morning/B afternoon with varying network speed: randomize or counterbalance method order within comparable conditions and record time. Timing remains an alternative explanation if left coupled.

Confounders and controls — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
New method uses better corpus and different queries while old method uses the original set.	For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.	Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Qualification: **Control**s address specified alternatives, not every possible unknown **confounder**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

New method uses better corpus and different queries while old method uses the original set.

Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Crossing, grouping and order-control probes

Confounding/control probe	Expected interpretation
Method, corpus and query set all change	Cannot isolate the method effect from this comparison
Only listed metadata match	Useful check; actual execution and unrecorded conditions may still differ
Healthy-white and spotted-blue only	Condition and background remain coupled
More of the same two cells	Missing cross-combinations remain missing
All four cells included	Improves this design; still inspect object grouping, sampling and other differences
Paper-only rule fits original records	Alternative explanation is possible; not proof a real model uses it
Same held-out leaf changes prediction with background	Shows sensitivity under that test, not complete reasoning
Repeated views counted as independent plants	Wrong unit count; retain object identity
Order/time changes with method	Randomize or counterbalance relevant order/time conditions and record them

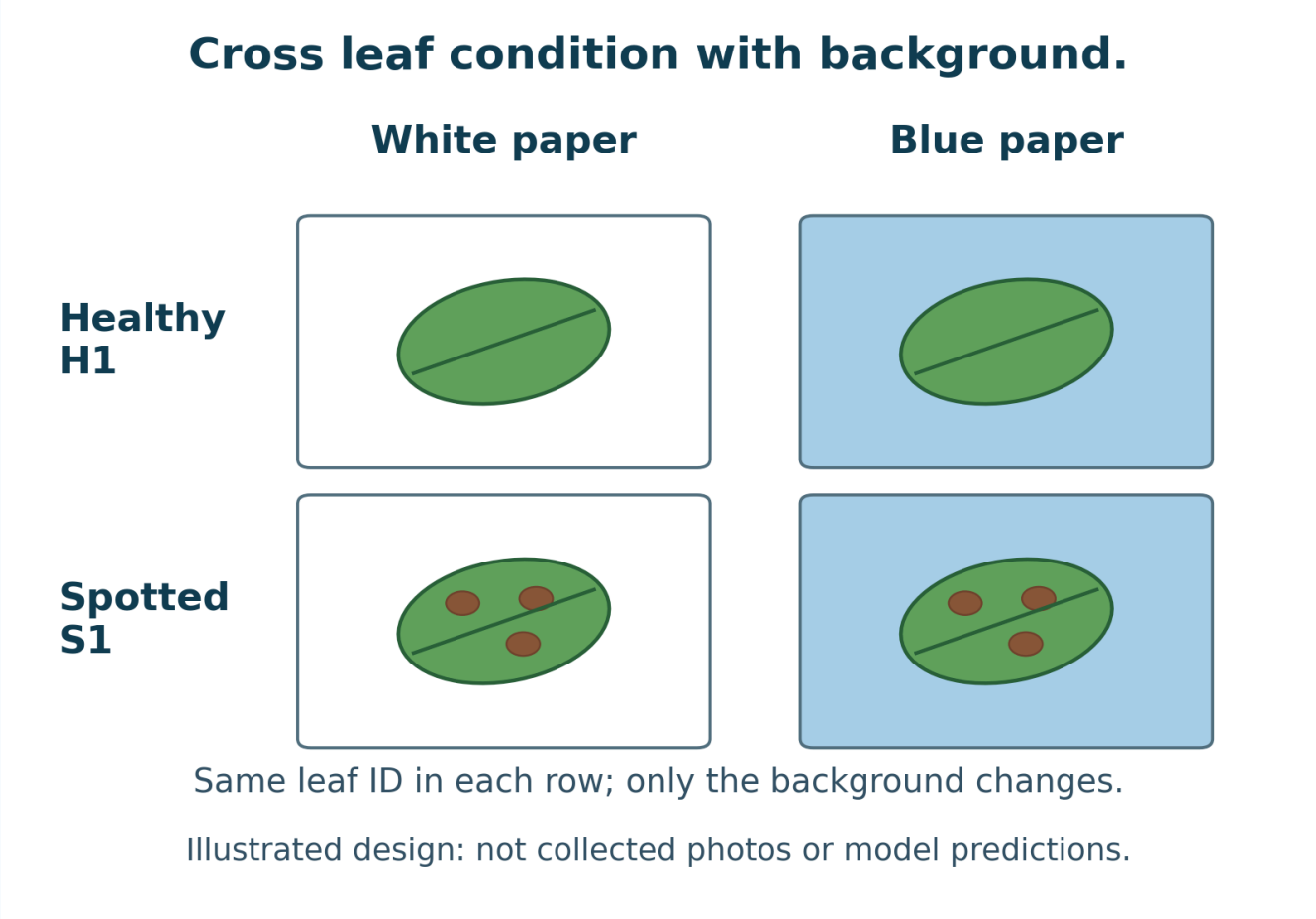
Separate leaf matrix, native illustration and toy-rule evidence

Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.

Goal is leaf condition, not paper colour. A proposed check uses both paper colours for each condition.

The picture is a separate leaf-background problem. If every healthy leaf is on white paper and every spotted leaf on blue, a rule using paper colour alone can fit those observed pairs. That possibility does not establish what a real model actually uses. Cross the two conditions and backgrounds, and inspect a paired test where the same held-out leaf changes background while its condition stays fixed.

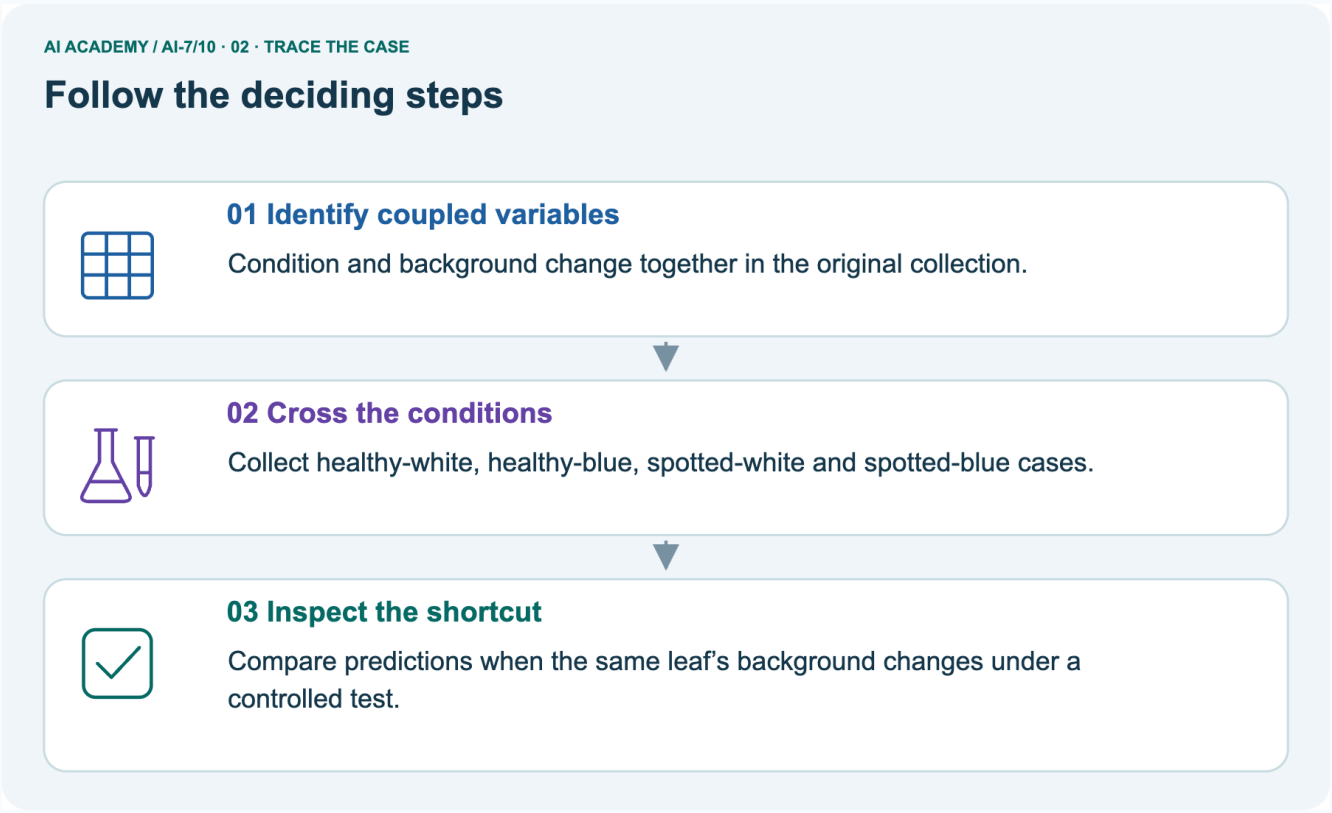
The native illustration shows healthy H1 and spotted S1 in both colours; these are repeated views of two objects. For a new-plant evaluation, keep all views of those objects out of training. A changed-background response can show sensitivity, not the model's complete internal reasoning or universal behavior.



Separate condition/background cell	Original collection	Crossed check	Explicit paper-only prediction
Healthy / white	Present	Included	Healthy: correct
Healthy / blue	Missing	Added	Spotted: incorrect

Separate condition/background cell	Original collection	Crossed check	Explicit paper-only prediction
Spotted / white	Missing	Added	Healthy: incorrect
Spotted / blue	Present	Included	Spotted: correct
Toy-rule total	2/2 correct on coupled records	2/4 correct on crossed records	Constructed alternative explanation, not actual model behavior
Object count	H1 and S1	Same H1/S1 in both backgrounds	Four records from two leaf objects

Picture trace and still-coupled-new-cases key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify coupled variables	Condition and background change together in the original collection.
2. Cross the conditions	Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.
3. Inspect the shortcut	Compare predictions when the same leaf's background changes under a controlled test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

Change: New cases still use only healthy-white and spotted-blue.

Prediction: The collection does not resolve the background confounding.

Reason: The same two variables remain coupled instead of being varied separately.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Confounder	A differing condition offering another explanation
Control	A design constraint preserving comparison conditions
Counterbalancing	Varying order to reduce systematic order differences

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Confounders and controls

The facts we will use

AI ACADEMY / AI-7/10

Cross class with paper background

	White paper	Blue paper
Healthy	Collect	Collect
Spotted	Collect	Collect

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/10

Confounders and controls

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify coupled variables

Condition and background change together in the original collection.



2. Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.



3. Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

CHANGE ONE THING

New cases still use only healthy-white and spotted-blue.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Confounder**s and **control**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Control**s address specified alternatives, not every possible unknown **confounder**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

Condition and background change together in the original collection. Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.

Compare **prediction**s when the same leaf's background changes under a **control**led test. Changed case: The collection does not resolve the background confounding.

The same two variables remain coupled instead of being varied separately.

Changed-case reasoning: The collection does not resolve the background confounding. The same two variables remain coupled instead of being varied separately.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main corpus and cases change with method; match the intended comparison or report a whole-system result without isolation. Picture crossing all four condition/background cells challenges a paper-only alternative, while repeating only the two old cells leaves coupling intact. Toy-rule 2/2 versus 2/4 is not an actual model result.

Selected A-always-morning/B-always-afternoon design needs randomized or counterbalanced method order under matched conditions and recorded timing. No automatic removal of every confounder or actual network measurement is implied.

Diagnose the current misconceptions

The worked result establishes every future case: **Control**s address specified alternatives, not every possible unknown **confounder**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

Teacher explanation: Counterbalance/randomise method order under matched conditions and record time; otherwise timing remains an alternative explanation.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **confounder** changes alongside the intended factor and can offer another explanation. **Control**s aim to isolate the proposed **mechanism**.

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

W2 Trace the worked case

New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

W3 Audit the limit

A report omits this qualification: **Control**s address specified alternatives, not every possible unknown **confounder**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: **Control**s address specified alternatives, not every possible unknown **confounder**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Control**s address specified alternatives, not every possible unknown **confounder**.

The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Qualification: **Control**s address specified alternatives, not every possible unknown **confounder**.
</details>

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

Teacher explanation: Counterbalance/randomise method order under matched conditions and record time; otherwise timing remains an alternative explanation.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Counterbalance/randomise method order under matched conditions and record time; otherwise timing remains an alternative explanation. Require intermediate reasoning and the stated scope.

Fresh retry: A uses unedited labels, B corrected labels. Expected reasoning: Label quality differs; use a shared declared label set to isolate procedure.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **confounder**s and **control**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **confounder**s and **control**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Sampling variability. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/11, ai-7/13, ai-7/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **confounder**s and **control**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Corpus and case differences confound the method effect. Rerun matched **input**s or report a whole-system comparison without isolation.

Check every learner before discussion: Use the worked case for **Confounder**s and **control**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Control**s address specified alternatives, not every possible unknown **confounder**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: New cases still use only healthy-white and spotted-blue. Require a **prediction** and a reason before comparing. Expected result: The collection does not resolve the background confounding.

Reason: The same two variables remain coupled instead of being varied separately.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

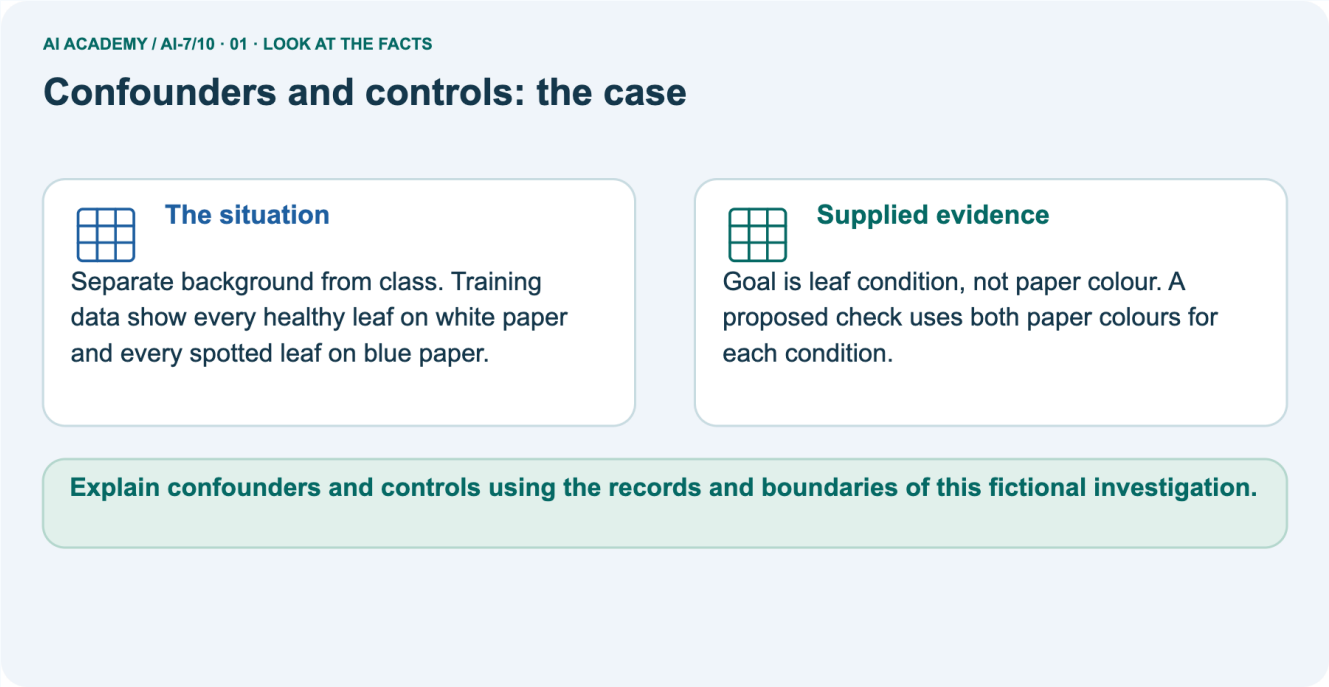
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Confounders and controls: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

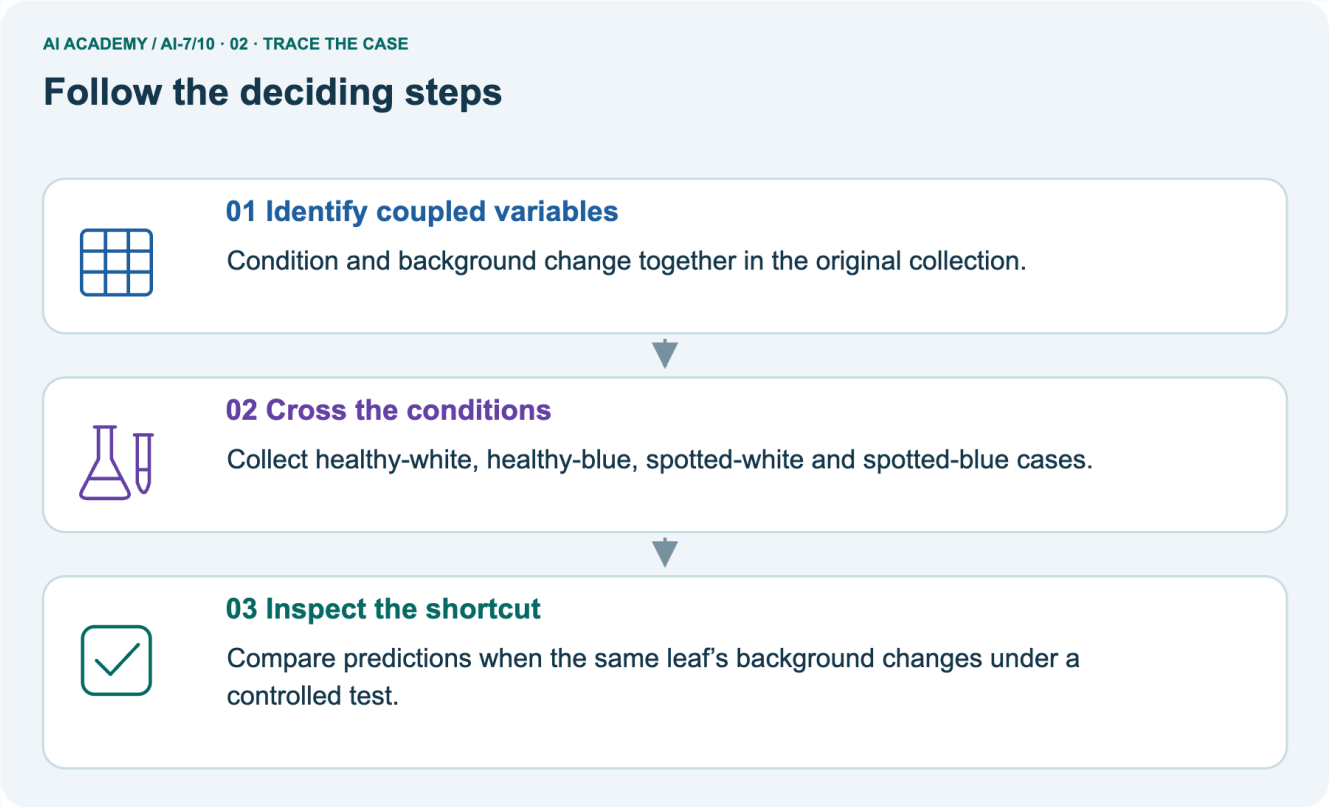
Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper. Supplied facts: Goal is leaf condition, not paper colour.

A proposed check uses both paper colours for each condition.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



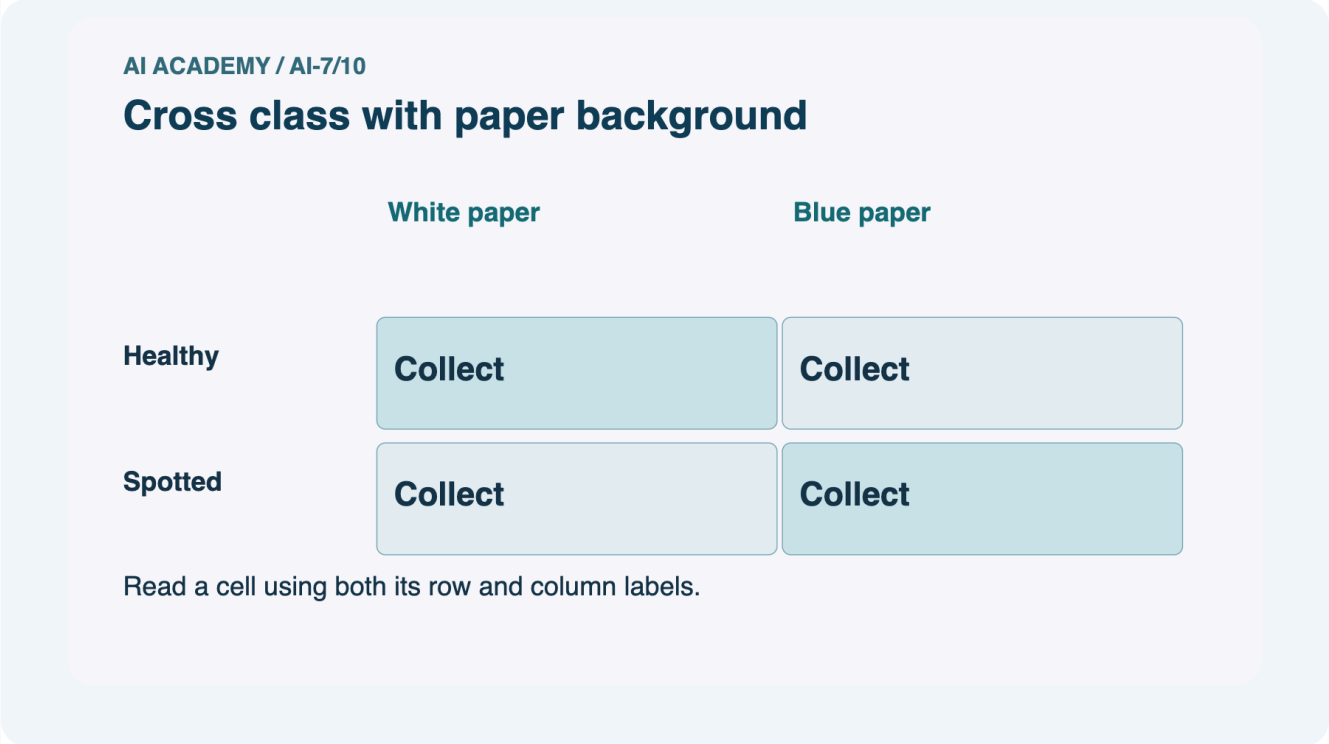
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify coupled variables: Condition and background change together in the original collection. → Cross the conditions: Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases. → Inspect the shortcut: Compare **prediction**s when the same leaf's background changes under a **control**led test.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Cross class with paper background



Follow the labelled picture with the matching explanation. Open the vector diagram

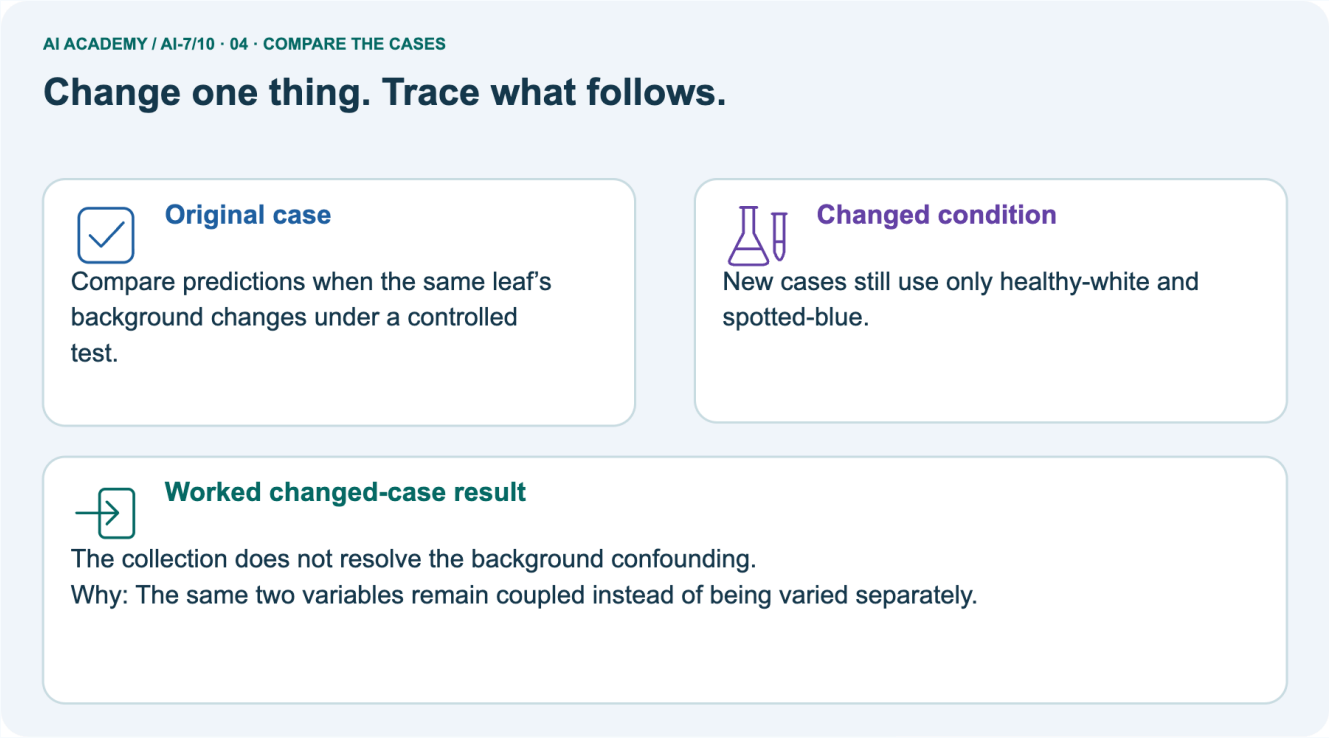
Cross class with paper background. Healthy: White paper = Collect, Blue paper = Collect. Spotted: White paper = Collect, Blue paper = Collect.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare **prediction**s when the same leaf's background changes under a **control**led test. Changed condition: New cases still use only healthy-white and spotted-blue.

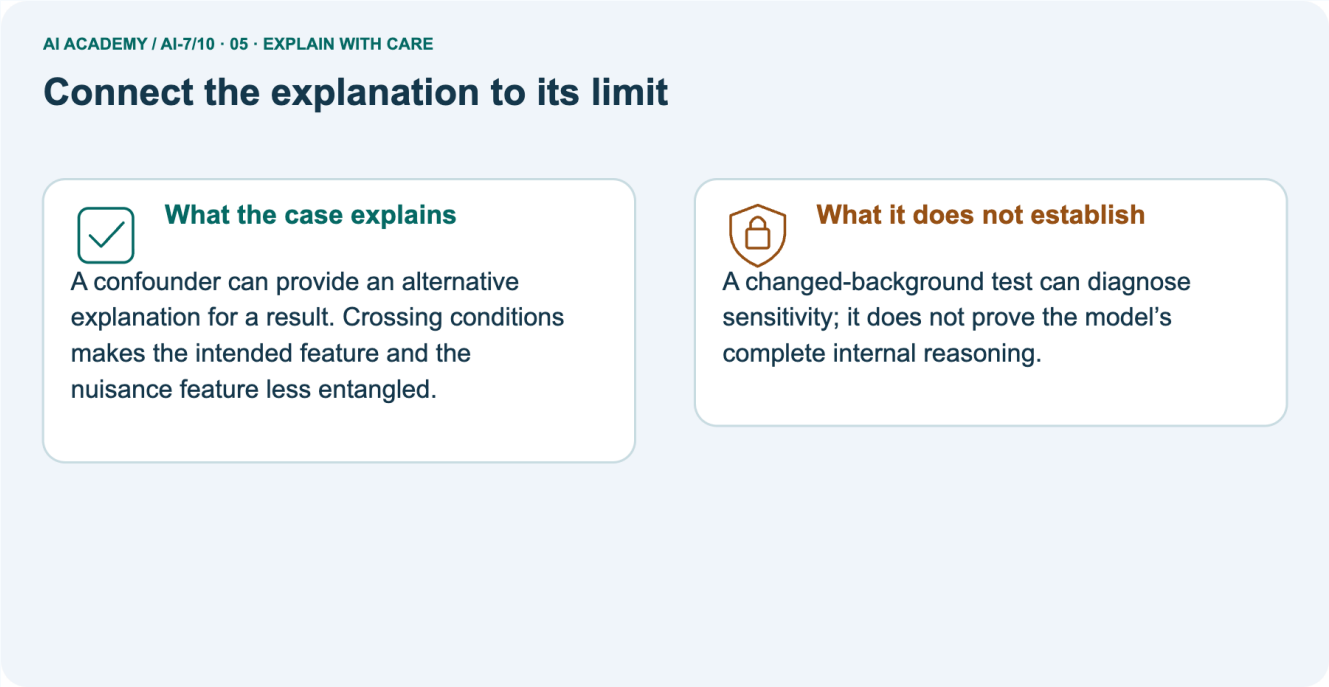
Expected result: The collection does not resolve the background confounding. Reason: The same two variables remain coupled instead of being varied separately.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The same two variables remain coupled instead of being varied separately.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

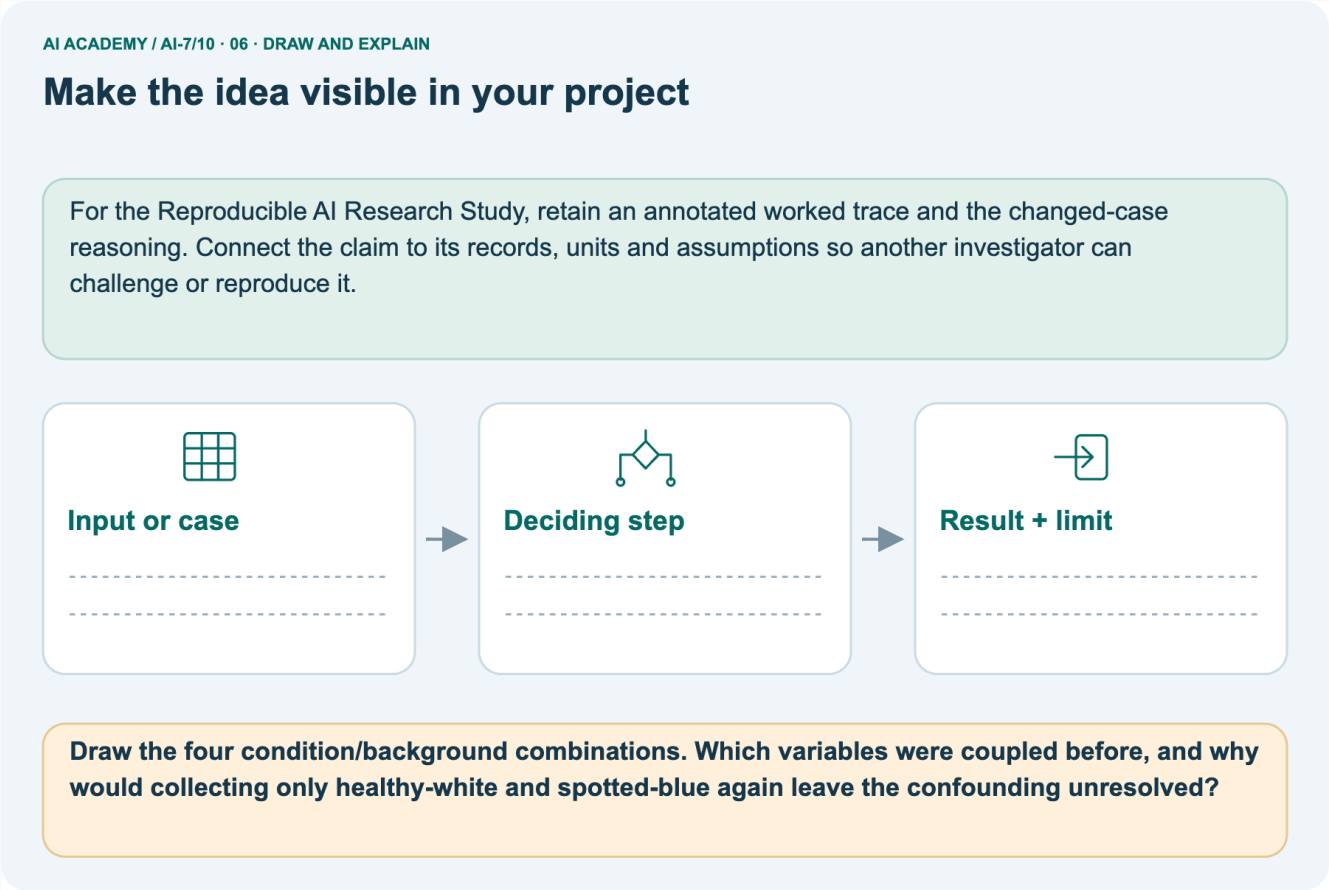
Limit: A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Condition and background change together in the original collection. Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.

Compare **prediction**s when the same leaf's background changes under a **control**led test. Changed case: The collection does not resolve the background confounding.

The same two variables remain coupled instead of being varied separately.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Confounders and controls: the case

02. Follow the deciding steps

03. Cross class with paper background

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.